

Virgo 120M Updated Fine-Tuning Dataset Expansion Plan

Goal

Objective	Transform Virgo from a Markdown-focused model into a general instruction-following assistant.
Target Dataset Size	250k-300k high-quality instruction pairs
Model Size	120M parameters

Current Dataset Strength

Area	Current Level
Markdown generation	Very Strong
Structured documents	Very Strong
Formatting control	Very Strong
Technical explanations	Strong
General conversation	Needs improvement
Reasoning	Needs improvement

Required Additions

Dataset Addition	Samples	Purpose
Natural Conversation	80,000	Teach assistant behavior, dialogue, helpful responses
General Instruction Following	60,000	Explain, summarize, rewrite, compare, create
Coding Dataset	25,000	Programming, debugging, algorithms
Science and Mathematics	20,000	Concepts, derivations, problem solving
Professional Writing	20,000	Emails, letters, reports, notices
Reasoning Dataset	15,000	Planning, logic, decision making
Formatting Control	10,000	Markdown, JSON, YAML, templates

Final Dataset Composition

Category	Samples
Existing Virgo Dataset	65,000
New Chat Dataset	80,000
Instruction Dataset	60,000
Coding	25,000
Science/Math	20,000

Category	Samples
Professional Writing	20,000
Reasoning	15,000
Formatting	10,000
Total Target	~295,000

Training Priority Order

Priority	Dataset
1	Natural Conversation (80k)
2	General Instructions (60k)
3	Coding (25k)
4	Professional Writing (20k)
5	Science/Math (20k)
6	Reasoning (15k)

Expected Virgo Improvement

Capability	Expected Result
Markdown generation	Excellent
Document creation	Excellent
Conversation ability	Strong
Instruction following	Strong
Coding assistance	Medium-Strong
Science explanations	Medium-Strong
Complex reasoning	Improved but limited by 120M size

Final recommendation: Maintain the existing Markdown dataset and add diverse conversation, instruction, coding, science, writing, and reasoning samples. A balanced 250k-300k dataset is a strong target for fine-tuning a 120M parameter model.